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Cordyceps militaris fruit body activates myeloid dendritic cells via a Dectin-1-mediated pathway

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Abstract

Cordyceps militaris, an entomopathogenic fungus, has been traditionally used in East Asian medicine. Recent research indicates that the fruit bodies of *C. militaris* are rich in bioactive compounds, such as polysaccharides and nucleosides, which may offer health benefits. However, the specific components responsible for its immunostimulatory effects and the mechanisms involved remain unclear. This study explored the immunomodulatory activity of a fruit body extract from *C. militaris*, named Ryukyu-kaso (RK), and examined the effect of the β -glucan receptor Dectin-1 on bone marrow-derived dendritic cells (BMDCs). Our results demonstrated that RK, which contains 1,3- β -glucan, effectively stimulated BMDCs to secrete pro-inflammatory and immunoregulatory cytokines and upregulated surface markers indicative of maturation and activation. Notably, these immunostimulatory effects were completely absent in BMDCs derived from Dectin-1-knockout mice, confirming that Dectin-1 is crucial for RK-induced immunomodulation. These findings provide new insights into the immunostimulatory mechanisms of *C. militaris* and underscore the potential of RK as a natural immunomodulatory agent for various therapeutic applications.

Keywords: *Cordyceps militaris*; Cytokine; Dectin-1; Myeloid dendritic cell; β -glucan.

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